

Question 25

Not yet answered

Marked out of 2.60

Flag question

$$\int_{-\infty}^{\infty} \sin(x + \pi) \delta(x) dx =$$

Select one:

- ☐ a. $\frac{1}{2}$
- ☐ b. 0
- ☒ c. $-\frac{1}{\sqrt{2}}$ ✓
- ☐ d. $\frac{1}{\sqrt{2}}$

Question 25

Not yet answered

Marked out of 2.00

Flag question

$$\int_{-\infty}^{\infty} \sin(x + \pi) \delta(x) dx =$$

Select one:

- ☐ a. $\frac{1}{2}$
- ☐ b. 0
- ☒ c. $-\frac{1}{\sqrt{2}}$ ✓
- ☐ d. $\frac{1}{\sqrt{2}}$

Question 24

Not yet answered

Marked out of 2.00

Flag question

A wide-sense stationary discrete-time process $X(nT_s)$, where T_s is the sampling constant and n can be any integer number. Then for every integer k with $C_{XX}(nT_s, nT_s + kT_s) = 1$ and $R_{XX}(kT_s) = 5$, then $R_{XX}(0) =$

Select one:

- ☐ a. 2
- ☒ b. 3 ✓
- ☐ c. 1
- ☐ d. 4

Question 23

Not yet answered

Marked out of 2.00

Flag question

For $z > 10$, we have $\delta(x) + u(x - 2) \equiv 1$

Select one:

- ☒ True ✓
- ☐ False

Question 22

Not yet answered
Marked out of 2.00

Flag question

For the random process $X(t) = \cos(wt)$, where w is a random variable uniformly distributed on the interval $[0, 1]$. Then which of the following is true

Select one:

- ☐ a. $E[X(t)] = \frac{\sin(t)}{t}$ and the process is not first order stationary
- ☐ b. $E[X(t)] = \frac{\sin(t)}{t}$ and the process is not first order stationary
- ☒ c. $E[X(t)] = \frac{\sin(t)}{t}$ and the process is first order stationary ✓
- ☐ d. $E[X(t)] = \frac{\sin(t)}{t}$ and the process is first order stationary

Question 21

Not yet answered

Marked out of 2.00

Flag question

Let $X(t)$ be a wide-sense stationary process. Suppose that $C_{xx}(t, t + \tau) = 4 \cos(\tau + 8\pi)$. Then the variance of the process is equal to

Select one:

- ☐ a. 3
- ☐ b. 4
- ☐ c. 5
- ☒ d. 2 ✓

Question 20

Not yet answered

Marked out of 2.00

Flag question

For the random process $X(t) = A \cos(wt)$ we have w is a positive constant and A is a random variable uniformly distributed on $[0, 1]$. Then

$$R_{xx}(t, t + \tau) = \frac{1}{3} \cos(wt) \cos(w(t + w\tau))$$

Select one:

- ☐ True
- ☒ False ✓

Question 19

Not yet answered

Marked out of 2.00

Flag question

Consider the random process $X(t) = A \cos(t)$, where A is a continuous random variable. If $E[A] = 1$ and $R_{xx}(t, t + \tau) = 5 \cos(t) \cos(t + \tau)$, then $\text{Var}(A) =$

Select one:

- ☐ a. 4
- ☒ b. 2 ✓
- ☐ c. 1
- ☐ d. 3

Question 13

Not yet answered

Marked out of 2.00

Flag question

Let $X \sim B(12, p)$ with $p = q$, then $\text{Var}(X) =$

Select one:

☐ a. 1☐ b. 4☐ c. 3☒ d. 2 ✓**Question 12**

Not yet answered

Marked out of 2.00

Flag question

Two independent and continuous random variables X and Y with pdfs

$$f_X(x) = u(x)e^{-x}$$

$$f_Y(y) = 3u(y)e^{-3y}$$

Then the density of $W = X + Y$ is

$$f_W(w) = \frac{3}{2}e^{-w}(1 - e^{-2w}) \text{ when } w \text{ is positive and zero otherwise}$$

Select one:

☐ True☒ False ✓**Question 11**

Not yet answered

Marked out of 2.00

Flag question

Let f_X be the pdf for some continuous random variable X . The pdf of the transformation $Y = e^X$ is

Select one:

☐ a.

$$f_Y(y) = f_X(\ln(y))$$

☐ b.

$$f_Y(y) = f_X(e^y)$$

☒ c.

$$f_Y(y) = \frac{1}{|y|} f_X(\ln(y)) \quad \checkmark$$

☐ d.

$$f_Y(y) = e^y f_X(e^y)$$

Question 10

Not yet answered

Marked out of 2.00

Flag question

Suppose that the jointly wide-sense stationary processes $X(t)$ and $Y(t)$ are independent with $E(X(t)) = 2$. If $R_{XY}(\tau) = 8$, then $E(Y(t)) =$

Select one:

☐ a. 3☐ b. 5☐ c. 4☒ d. 2 ✓

Question 9

Not yet answered
Marked out of 2.00

Flag question

Consider two wide-sense stationary processes $X(t)$ and $Y(t)$ with cross-correlation $R_{xy}(t, t + \tau) = 0$. Then $X(t)$ and $Y(t)$ are jointly wide-sense stationary but are NOT orthogonal

Select one:

- ☐ a. True
☒ b. False

Question 8

Not yet answered
Marked out of 2.00

Flag question

For some random process $X(t)$ with $E[X^2(t)] = b + \sin(t)$. If the average power $P_{xx} = 2$; then the value of the positive constant b equals ✓

Select one:

- ☐ a. 1
☐ b. 2
☐ c. 4
☒ d. 3 ✓

Question 7

Not yet answered
Marked out of 2.00

Flag question

A discrete random variable X with range $\{0, 1\}$ and density

$$f_X(x) = \frac{1}{3}\delta(x) + \frac{2}{3}\delta(x - 1). \text{ Then } \Pr(X = 1) =$$

Select one:

- ☐ a. $\frac{1}{3}$
☐ b. 0
☐ c. 1
☒ d. $\frac{2}{3}$ ✓

Question 6

Not yet answered
Marked out of 2.00

Flag question

For a random variable $X \sim N(\mu, \sigma^2)$, if $\Pr\left(\frac{x-\mu}{\sigma} < 0\right) = 0.5$, then $x = \mu$

Select one:

- ☒ a. True ✓
☐ b. False

Question 5

Not yet answered

Marked out of 2.00

Flag question

For a random variable $Z \sim N(0, 1)$ we have $P_r(Z < -a) \neq P_r(Z > a)$ for any constant $a > 0$

Select one:

- ☐ True
- ☒ False ✓

Question 4

Not yet answered

Marked out of 2.00

Flag question

For a random process with auto-correlation function $B_{XX}(\tau) = 3 + be^{-|\tau|}$, where b is a positive constant. The process is stationary ergodic with no periodic components. Then $E[X]^3 =$

Select one:

- ☐ a. 2
- ☐ b. 5
- ☐ c. 4
- ☒ d. 3 ✓

Question 3

Not yet answered

Marked out of 2.00

Flag question

For random variables $X \sim N(1, 1)$ and $Z \sim N(0, 1)$; we have $P_r(X < 2) =$

Select one:

- ☒ a. $P_r(Z < 2)$ ✓
- ☐ b. $P_r(Z < 1)$
- ☐ c. $P_r(Z < 3)$
- ☐ d. $P_r(Z < 4)$

Question 2

Not yet answered

Marked out of 2.00

Flag question

Let $X \sim \text{Poisson}(k)$ with pdf

$$P_r(X = x) = \frac{(0.5)^x e^{-0.5}}{e^{0.5} \cdot x! \cdot 2^x}, x = 0, 1, 2, \dots$$

then $k = 0.5$

Select one:

- ☐ True
- ☒ False ✓

Question 1

Not yet answered

Marked out of 2.00

Flag question

Consider a random variable X with $E(X^2) = 10$. If X is orthogonal with $Y = X - 2$, then $E(X) = 10$

Select one:

- ☐ True
- ☒ False ✓

Question 18

Not yet answered

Marked out of 2.00

Flag question

For the random process $X(t) = A^2 \cos(\omega t)$ we have ω is a positive constant and A is a random variable uniformly distributed on $[-5, 5]$. Then $E[X(t)] =$

Select one:

- ☒ a. $\frac{25}{3} \cos(\omega t)$ ✓
- ☐ b. $\cos(\omega t)$
- ☐ c. $\frac{1}{4} \cos(\omega t)$
- ☐ d. 0

Question 17

Not yet answered

Marked out of 2.00

Flag question

A random process with auto-correlation function $R_{XX}(r) = b + e^{-|r|}$. The process is stationary ergodic with no periodic components. If $E[X^2(t)] = 1$, then the value of the positive constant b equals

Select one:

- ☐ a. 3
- ☐ b. 1
- ☒ c. 0 ✓
- ☐ d. 2

Question 15

Not yet answered

Marked out of 2.00

Flag question

Let $X \sim B(n, p)$ with pdf

$$P_r(X = x) = \left(\frac{1}{2}\right)^{100} \binom{100}{x}, \quad x = 0, 1, 2, \dots, 100$$

then $p = 0.1$

Select one:

- ☐ True
- ☒ False ✓

Question 16

Not yet answered

Marked out of 2.00

Flag question

A random process $X(t)$ with power density spectrum $p_{XX}(\omega) = 2\pi(\delta(\omega) + \delta(\omega - 1))$. The average power P_{XX} equals

Select one:

- ☐ a. 4
- ☐ b. 1
- ☒ c. 2 ✓
- ☐ d. 3

Question 14

Not yet answered

Marked out of 2.00

Flag question

Random variable Z is standard normally distributed with $P(Z < \alpha) = 0.97725$, then $\alpha = 1$

Select one:

- ☐ True
- ☒ False ✓

Let X and Y be two random variables with pdf's...

$$f_X(x) = u(x) e^{-x}$$

$$f_Y(y) = 2u(y) e^{-2y}$$

The pdf for $W = X + Y$ is

Select one:

- ☒ a. $f_W(w) = 2e^{-w}(1 - e^{-w})$ when w is positive and zero otherwise ✓
- ☐ b. $f_W(w) = 2e^{-w}(1 - e^{-w})$ when w is positive and zero otherwise
- ☐ c. $f_W(w) = (|w|w|(1-w)|w|w|)$ when w is positive and zero otherwise

X be a continuous random variable with pdf f_X . The pdf of the transformation $Y = 1 - X$ is

Select one:

- ☐ a. $f_Y(y) = f_X(1 - y)$
- ☒ b. $f_Y(y) = f_X(1 - y)$ ✓
- ☐ c. $f_Y(y) = f_X(1 - y)$
- ☐ d. $f_Y(y) = f_X(1 - y)$

For a random variable $Y \sim N(\mu, 2)$, if $\Pr(Y > 0) = 0.5$, then Y is standard normally distributed

Select one:

- ☐ True
- ☒ False ✓

For two orthogonal random variables X and Y with $E(X) = E(Y)$ and $C_{XY} = -9$; then $|E(X)| = 3$

Select one:

- ☒ True ✓
- ☐ False

For a random process with auto-correlation function $R_{XX}(\tau) = 2 + e^{-|\tau|}$. The process is stationary ergodic with no periodic components. We have a?

Select one:

- ☐ a. 2
- ☒ b. 1 ✓
- ☐ c. 4
- ☐ d. 3

For two orthogonal random variables X and Y with $E[X] = E[Y] = 2$ and $C_{xy} = -4$; then $E[XY] = ?$

Select one:

- ☒ a. 0 ✓
- ☐ b. 4
- ☐ c. 2
- ☐ d. -4

For a random process with auto-correlation function $R_{xx}(\tau) = 2 + e^{-|\tau|}$. The process is stationary ergodic with no periodic components. We have a?

Select one:

- ☐ a. 2
- ☒ b. 1 ✓
- ☐ c. 4
- ☐ d. 3

For a wide-sense stationary process $X(t)$, assume that $C_{xx}(t, t + \tau) = \cos(\tau) + 4$. Then the variance of the process equals

Select one:

- ☐ a. 2
 - ☐ b. 5
 - ☐ c. 3
 - ☒ d. 4 ✓
- 

For two orthogonal random variables X and Y with $E[X] = E[Y] = 2$ and $C_{xy} = -4$; then $E[XY] = ?$

Select one:

- ☒ a. 0 ✓
- ☐ b. 4
- ☐ c. 2
- ☐ d. -4

For a random process with auto-correlation function $R_{xx}(\tau) = 2 + e^{-|\tau|}$. The process is stationary ergodic with no periodic components. We have a?

Select one:

- ☐ a. 2
- ☒ b. 1 ✓
- ☐ c. 4
- ☐ d. 3

For a wide-sense stationary process $X(t)$, assume that $C_{xx}(t, t + \tau) = \cos(\tau) + 4$. Then the variance of the process equals

Select one:

- ☐ a. 2
 - ☐ b. 5
 - ☐ c. 3
 - ☒ d. 4 ✓
- 

Let $X(t)$ be a wide-sense stationary process.
Suppose that $C_{XX}(t, t + \tau) = 2\cos(\tau + 8\pi)$.
Then the variance of the process is equal to

Select one:

- ☐ a. 5
- ☐ b. 4
- ☒ c. 2 ✓
- ☐ d. 3

Let $X \sim B(8, p)$ with $p = q$,
then $\text{Var}(X) =$

Select one:

- ☒ a. 2 ✓
- ☐ b. 1
- ☐ c. 3
- ☐ d. 4

A random process with auto-correlation function $R_{xx}(\tau) = b + e^{-|\tau|}$. The process is stationary ergodic with no periodic $E[X^2(t)] = 2$, then the value of the positive constant b equals

Select one:

- ☐ a. 2
- ☐ b. 3
- ☐ c. 0
- ☒ d. 1 ✓

Let $X(t)$ be a wide-sense stationary process. Suppose that $C_{xx}(t, t + \tau) = 2 \cos(\tau + 8\pi)$. Then the variance of the process is equal to

Select one:

- ☐ a. 5
- ☐ b. 4
- ☒ c. 2 ✓
- ☐ d. 3

Consider a random variable X with $E(X^2) = 10$. If X is orthogonal with $Y = X - 2$, then $E(X) = 10$

Select one:

- ☐ True
- ☒ False ✓

For $x > 10$, we have $\delta(x) + u(x - 2) = 1$

Select one:

- ☒ True ✓
- ☐ False

For the random process $X(t) = \cos(\omega t)$ where ω is a random variable uniformly distributed on the interval $[0, 1]$. Then which of the following is true

Select one:

- ☐ a. $E[X(t)] = \frac{\sin(t)}{t}$ and the process is not first order stationary
- ☐ b. $E[X(t)] = \frac{\sin(t)}{t}$ and the process is first order stationary
- ☒ c. $E[X(t)] = \frac{\sin(t)}{t}$ and the process is not first order stationary
- ☐ d. $E[X(t)] = \frac{1 - \cos(t)}{t}$ and the process is not first order stationary

Consider two wide-sense stationary processes $X(t)$ and $Y(t)$ with cross-correlation $R_{XY}(t, t + \tau) = 0$. Then $X(t)$ and $Y(t)$ are jointly wide-sense stationary but are NOT orthogonal

Select one:

- ☐ True
- ☒ False

For a random process with auto-correlation function $R_{xx}(\tau) = 2 + be^{-|b\tau|}$, where b is a positive constant. The process is stationary ergodic with no periodic components. Then $\text{Var}(X) = ?$

Select one:

- ☐ a. 4
- ☐ b. 2
- ☐ c. 5
- ☒ d. 3 ✓

For $x > 10$, we have
 $\delta(x) + u(x - 2) = 1$

Select one:

- ☒ True ✓
- ☐ False

For some random process $X(t)$ with $E(X^2(t)) = b + \sin(t)$. If the average power $P_{xx} = 1$; then the value of the positive constant b equals

Select one:

- ☐ a. 4
 - ☐ b. 2
 - ☒ c. 1 ✓
 - ☐ d. 3
- $$\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} E[X^2(t)] dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} (b + \sin(t)) dt = \lim_{T \rightarrow \infty} \frac{bT}{T} \left[bT - 2 \cos\left(\frac{T}{2}\right) \right] = b$$